CIVICCONNECT
PRODUCT ERROR / DEBUGGING DOCUMENT

PED v3.0 | Release Candidate - M3

Submission and release evidence for the CivicConnect M3 milestone.

VERSION	DATE	CLASSIFICATION	TAG / COMMIT
3.0	2026-10-16	CONDITIONALLY READY	a1e189b

Repository: github.com/Armand-JV/SEN381-Project

Prepared by the CivicConnect project team

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Document note

This PDF presents the supplied Markdown as a navigable report. Requirement claims, test results, limitations, and release conditions are reproduced from the source document.

Release Candidate — M3

Version: 3.0 **Date:** 2026-10-16 **Repository:** https://github.com/Armand-JV/SEN381-Project **Tag/Commit:** `a1e189b` (Riegardt Phase 6: management slice, release review and readiness evidence — #138) **PED v3.0:** `docs/Milestone 3/PEDv3.0.md` **RTM:** `docs/Milestone 3/RTM-M3.md` **CI:** `.github/workflows/test.yml`

Team members

Role	Person	Rubric areas	Slices owned
Quality & Verification Lead	Ditheto	A (M2 conformance), D (black-box/integration/system), E (RTM/defects/traceability)	Requester slice FR-001–006
Construction, CI & Security Lead	Armand	B (collaborative construction), C (CI & quality gates)	Staff slice FR-007–013, NFR-017–019
Release, Performance & Operations Lead	Riegardt	F (performance/staging), G (release review, residual risk)	Management slice FR-014–016, NFR-020, NFR-024

Submission date: 2026-10-16 **Classification: CONDITIONALLY READY** (see Section 10)

1. Release Candidate Scope and Submission Information

1.1 In-scope requirements (frozen)

The M3 release candidate covers the agreed core scope. Deferred items are recorded in the Technical Debt Register with reasons.

ID	Requirement	Owner	RTM status	Evidence
FR-001	Submit validated service request	Ditheto	Implemented	`docs/Tests/requester-slice.md`; `RequesterSliceTests.cs`
FR-002	Category selection and category-specific validation	Ditheto	Implemented	`docs/Tests/issue_27_observer_test_evidence.md`; `CategoryValidationContextTests.cs`
FR-003	Unique request reference	Ditheto	Verified	Numeric `Request.Id` accepted as reference (Riegardt decision)
FR-004	Requester views own request and status	Ditheto	Verified	`RequesterSliceTests.cs`, `GetMyRequests_*`
FR-005	Requester views status history timeline	Ditheto	Verified	`RequesterSliceTests.GetHistory_*`
FR-006	Requester feedback on completed request	Ditheto	Implemented	`RequesterSliceTests.Feedback_*`; `FeedbackController.cs`
FR-007	Staff view relevant requests	Armand	Implemented	`docs/Tests/github_issues_22-25_test_evidence.md`; `RequestControllerTests.GetStaffRequests_*`
FR-008	Filter/search/sort requests	Armand	Implemented	`GetStaffRequests_CombinesFiltersAndPaging`;

ID	Requirement	Owner	RTM status	Evidence
FR-009	Staff view full request details	Armand	Implemented	`RequestControllerTests.GetRequest_SeededRequestIncludesComment`
FR-010	Assign request to staff	Armand	Verified	`RequestControllerTests.AssignRequest_*`
FR-011	Public and internal comments	Armand	Implemented	`RequestControllerTests.AddComment_*`; full matrix not covered
FR-012	State update transitions	Armand	Implemented	Partial; full matrix not covered
FR-013	Staff scope restriction	Armand	Implemented	`GetRequest_StaffOutsideAssignment_ReturnsForbidden`
NFR-017	Authentication	Armand	Implemented	`Program.cs` JWT bearer; test handler limitation (RR-SEC-04)
NFR-018	Role-based authorization	Armand	Implemented	Role attributes on controllers; `AccessControlService`
NFR-019	Object-level access control	Armand	Implemented	Requester ownership verified; staff-assign scope tested
FR-014	Management dashboard	Riegardt	Verified	`DashboardSummaryTests.cs`; `docs/Riegardt-Decisions.md` M3-PERF-01
FR-015	Management oversight: stalled + reassign	Riegardt	Verified	`ManagementTests.cs`; `docs/Riegardt-Decisions.md` FR-015
FR-016	Category administration	Riegardt	Verified	`ManagementTests.cs`; `docs/Riegardt-Decisions.md` FR-016
NFR-020	Audit trail	Riegardt	Verified	`AuditLogTests.cs`; `docs/Riegardt-Decisions.md` NFR-020
NFR-024	Responsive everyday operations	Riegardt	Verified	TC-P-01: p95 23.71ms (target 3s) — PASS

1.2 Deferred requirements

ID	Requirement	Reason	Debt link
FR-008 search	Full-text search on requests	Deferred by team decision	`docs/Technical-Debt-Register.md` TD-10
Dashboard age buckets	Request age bucketing in summary	Deferred by user decision	`docs/Technical-Debt-Register.md` TD-10
Outbox/retry for observers	Durable notification delivery	Deferred beyond M3 (accepted)	`docs/Technical-Debt-Register.md` TD-03

1.3 Submission references

Artefact	Location
Source code	`src/` (4 projects: Domain, Application, Web, Tests + Perf)
CI workflow	`.github/workflows/test.yml`
RTM	`docs/RTM-M3.md`
Test evidence	`docs/Tests/`
Registers (defects, debt, risks, AI)	`docs/` root
Staging scripts	`scripts/staging-e2e.ps1`, `scripts/staging-smoke.ps1`
Performance harness	`src/CivicConnect.Perf/`

2. M2 Baseline → Implementation Conformance Table

Adapted from `docs/Milestone 3/CivicConnect M2 ADR Transition Tables (M2 to M3).md`.

ADR	M2 decision	Conformance	Evidence	Status
ADR-01	Layered: Domain / Application / Web / Tests	Partial — `IRequestRepository`/`ICategoryRepository` interfaces exist and implemented; `RequestsController` uses `DbContext` directly; Application service layer	`src/CivicConnect.Application/`, `src/CivicConnect.Web/Controllers/RequestsController.cs`	Significant deviation (Section 3 / ADR-01)
ADR-02	PostgreSQL + EF Core, code-first migrations	Conformant on schema	`CivicConnectDbContext.cs`, `Migrations/`, `db_schema.sql`	Refined — integrity gaps addressed (cascade restrict, concurrency token in Riegardt decisions)
ADR-03	.NET 10 / ASP.NET Core / Swagger / xUnit, GitHub Actions	Conformant on versions	`Directory.Build.props`, CI workflow, project files	Refined — config, CI gates extended
ADR-04	Observer pattern for state changes	Interface + `RequestStateNotifier` exist; concrete `LogRequestStateObserver` now registered and tested	`RequestStateNotifier.cs`, `Observer/` in Application	Completed — concrete observer registered (Issue #27)
ADR-05	Strategy pattern for category validation	Interface + `CategoryValidationContext` exist; `Category1ValidationStrategy` now registered	`Strategy/`, `Program.cs` registrations	Completed — concrete strategy registered
ADR-06	REST API via ASP.NET Core controllers	Endpoints for all slices	`Controllers/`, README API list	Updated — JWT auth, role attributes, DTOs added

Significant deviations requiring ADR records

1. **Repository abstraction bypass (ADR-01):** `RequestsController` uses `CivicConnectDbContext` directly. Accepted simplification, documented as TD-01. No repository implementations added.
2. **Production JWT validation not exercised by tests:** `TestHeaderAuthenticationHandler` proves controller wiring only (RR-SEC-04). Production JWT validation is not tested.

3. **API path version:** PED v2 planned `/api/v1/`; implementation uses `/api`. Accepted change, documented in RTM evidence limits.
4. **State pattern:** PED v2 proposed State objects; implementation uses central `RequestStateTransitionPolicy` table. Accepted change, tested.

3. Controlled Changes

3.1 ADR-01 change record — DbContext directly in controller

Stage	Content
Original decision (M2)	Repository interfaces in Application; thin controllers
Implementation evidence	`IRequestRepository`/`ICategoryRepository` declared in `CivicConnect.Application/Interfaces/` but no implementations exist; nothing uses them
Deviation found	`RequestsController` injects `CivicConnectDbContext` directly; business logic (validation, state change, history, notification) in controller
Revised decision	Accept DbContext-as-UoW as a deliberate simplification (Option B from M2 ADR document). Delete unused repository interfaces or implement repositories per ADR-M3-02 if testability demands.
Recorded as	Technical Debt TD-01
Status	Accepted

3.2 State-pattern change — central transition table

Stage	Content
Original decision (M2)	State pattern objects
Implementation evidence	`RequestStateTransitionPolicy.cs` — static `PermittedTransitions` dictionary
Revised decision	Central transition table is simpler and equally testable
Consequence	RTM design-decision column re-pointed; behavior tested via `RequestTransitionPolicyTests` and controller tests
Status	Accepted; PED conformance row updated

3.3 Authentication/authorization change

Stage	Content
Original decision (M2)	RBAC with real identity provider
Implementation evidence	JWT bearer authentication in `Program.cs`; `TestHeaderAuthenticationHandler` for test environments only; role attributes `[Authorize(Roles = "STAFF,MANAGER")]`
Revised decision	JWT bearer for production; test-header scheme only in `Testing` environment (Development + `DemoAuth:Enabled`)

Stage	Content
Consequence	Controller authorization wiring verified; production JWT validation not exercised (RR-SEC-04)
Status	Accepted with residual risk

4. Construction and Configuration

4.1 Project structure

```
src/
■■■ CivicConnect.Domain/       — Entities, Enums, POCOs
■■■ CivicConnect.Application/  — Policies, Services, Patterns (Observer, Strategy)
■■■ CivicConnect.Web/          — Controllers, DbContext, Migrations, SPA
■■■ CivicConnect.Tests/        — Unit + integration tests
■■■ CivicConnect.Perf/         — NBomber load tests
```

Dependencies flow Web → Application → Domain. Web also references Domain directly. Tests reference all three.

4.2 Domain layer

Entities (`src/CivicConnect.Domain/Entities/`):

- `Request` — `Id`, `Title`, `Description`, `CategoryId`, `State`, `CreatedAt`, `RequesterId`, `AssigneeId`, navigation collections
- `Category` — `Id`, `Name`, `Description`
- `Comment` — `Id`, `RequestId`, `Text`, `CreatedAt`, `CreatedBy`, `IsInternal`
- `StatusHistory` — `Id`, `RequestId`, `PreviousState`, `NewState`, `ChangedAt`, `ChangedBy`, `Reason`
- `AuditLog` — `Id`, `Actor`, `Action`, `EntityType`, `EntityId`, `BeforeSummary`, `AfterSummary`, `Timestamp`, `CorrelationId`
- `AuditEntry` — deprecated; side-by-side with `AuditLog` (TD-04)
- `Feedback` — `Id`, `RequestId`, `Rating` (1–5), `Comment` (≤2000), `CreatedAt`, `CreatedBy`

`RequestState` enum: `Open, Assigned, InProgress, Resolved, Closed, Cancelled, Rejected`

4.3 Application layer

Policies:

- `RequestStateTransitionPolicy` (`src/CivicConnect.Application/Policies/`) — central transition table; `RequiresReason` for Rejected/Cancelled/Closed
- Permitted transitions:
- Open → Assigned, Rejected, Cancelled
- Assigned → InProgress, Rejected, Cancelled
- InProgress → Resolved, Cancelled
- Resolved → Closed
- Closed → (terminal)

- Cancelled → (terminal)
- Rejected → (terminal)

Patterns:

- Observer: `IRequestStateObserver` + `RequestStateNotifier` (fans out sequentially; failures isolated)
- Strategy: `CategoryValidationContext` selects `ICategoryValidationStrategy` by category ID; fallback non-empty title/description validation

4.4 Web/config layer

DI registrations (`Program.cs`):

```
- DbContext<CivicConnectDbContext> (Npgsql)
- CategoryValidationContext (scoped)
- ICategoryValidationStrategy → Category1ValidationStrategy (scoped)
- IRequestStateObserver → LogRequestStateObserver (scoped)
- RequestStateNotifier (scoped)
- IAccessControlService → AccessControlService (scoped)
```

Environment configuration:

- `EnvLoader` loads `.env` from content/current/base directories upward without overwriting existing process variables
- DB precedence: `DB_*` vars → `ConnectionStrings:DefaultConnection` → local PostgreSQL defaults
- JWT from `JWT_ISSUER`, `JWT_AUDIENCE`, `JWT_SECRET`

API endpoints:

Endpoint	Roles	Description
`POST /api/requests`	Auth	Submit request (validated, requester from principal)
`GET /api/requests/{id}`	REQ own / STF / MGR	Request detail; internal comments hidden from requesters
`GET /api/requests/mine`	REQ	Own requests, paged
`GET /api/requests/{id}/history`	owner / STF / MGR	Status history timeline
`PUT /api/requests/{id}/state`	STF / MGR	State transition (reason required for Reject/Cancel)
`PUT /api/requests/{id}/assign`	STF / MGR	Assign to staff, Open→Assigned
`POST /api/requests/{id}/comments`	REQ public/own / STF	Public/internal comments
`POST /api/requests/{id}/feedback`	REQ owner, Resolved/Closed	Rating 1–5, one per request
`GET /api/categories`	Auth	Seeded category list
`GET /api/staff/requests`	STF / MGR	Queue with filters + paging
`GET /api/reports/requests-summary`	MGR	Counts by state/category
`GET /api/audit`	MGR	Append-only audit trail

Endpoint	Roles	Description
`GET /api/management/stalled`	MGR	Open/Assigned older than N days
`PUT /api/management/requests/{id}/reassign`	MGR	Reassign with history + audit
`POST /api/management/categories`	MGR	Create category
`PUT /api/management/categories/{id}`	MGR	Rename category
`GET /health`	Anonymous	Database connectivity
`GET /api/version`	Anonymous	Build/commit/environment

Swagger: Enabled in Development and Staging (`Program.cs`); never Production.

Frontend: React 19 + Vite SPA (`src/CivicConnect.Web/ClientApp/`), built output served from `wwwroot` with SPA fallback.

5. CI and Automated Verification

5.1 CI workflow

`.github/workflows/test.yml` runs on push/PR to `main`:

1. Checkout repository
2. Setup Node.js 22 (for frontend build)
3. Build frontend (`npm ci` + `npm run build`)
4. Setup .NET 10
5. Restore dependencies
6. Build solution (Release, warnings-as-errors)
7. Install EF tool
8. Apply database migrations (PostgreSQL service on port 5432)
9. Verify formatting (`dotnet format --verify-no-changes`)
10. Check vulnerable packages (`scripts/check-vulnerabilities.py`)
11. Validate coverage script self-test
12. Test solution with TRX + XPlat coverage (Cobertura format)
13. Summarise Cobertura coverage (`scripts/coverage-summary.py`)
14. Upload test results artifact

5.2 Quality gates

Gate	Mechanism	Blocks merge
Build	`dotnet build` (warnings-as-errors)	Yes

Gate	Mechanism	Blocks merge
Format	`dotnet format --verify-no-changes`	Yes
Tests	`dotnet test` (all must pass)	Yes
Vulnerabilities	`scripts/check-vulnerabilities.py` (rejects high/critical)	Yes
PostgreSQL	Migration applies, Requests table confirmed	Yes

5.3 Branch protection

`.github/Main Branch protection.json`:

- Requires 2 non-author approvals
- Dismiss stale reviews on push
- Require last push approval
- Required linear history
- Copilot code review on push
- No force pushes to `main`

5.4 Coverage interpretation

CI collects Coverlet Cobertura output during `dotnet test`. `scripts/coverage-summary.py` reports line/branch totals per report. CI does **not** enforce an arbitrary coverage percentage threshold.

Coverage counts instrumented lines/branches executed. It does **not** prove:

- Assertions are meaningful
- Acceptance criteria are satisfied
- Authorization boundaries are complete
- Untested production configuration is safe

Requirement status comes from the RTM, not coverage percentages. See `docs/Milestone 3/PED-Section-5-CI-and-Coverage.md`.

5.5 Critical-path gaps

Per `docs/Milestone 3/PED-Section-5-CI-and-Coverage.md`:

- Staff object-level request scope: not blocked or tested (FR-007, NFR-019)
- Production authentication: issuer/audience/signature not exercised (NFR-017, NFR-018)
- State workflow: full transition matrix not fully covered (FR-011)
- Resolution journey: requester-visible state after staff resolution lacks E2E assertion (FR-013)
- Detail and resolution evidence: exhaustive field assertions absent (FR-009, FR-012)

5.6 Static/security findings

Canonical register: `docs/CI/M3-CI-07-findings.md`. See Section 9 for discharge.

ID	Tool	Severity	Finding	Disposition
FIND-001	dotnet format	Medium	Whitespace violations	Fixed
FIND-002	CA1707	Warning	Underscored test names	Mitigated (.editorconfig exception)
FIND-003	CA1861	Warning	Constant array in migration	Mitigated (.editorconfig exception)
FIND-004	CA1870	Warning	Non-cached SearchValues	Fixed
FIND-005	dotnet list package	Informational	No vulnerable packages	Accepted (monitoring, RR-SEC-02)
FIND-006	Test harness	High	JWT test fix failures	Fixed
FIND-007	Test isolation	Medium	EF context cross-scope	Fixed
FIND-008	History scan	High	Historical credential in Git	Mitigated (RR-SEC-01, TD-SEC-01)
FIND-009	gh settings	Medium	Unrelated .gitignore change	Out of scope
FIND-010	GitHub settings	Medium	Missing security settings	Accepted (RR-SEC-03)

5.7 Verification evidence

Check	Result	Date
Local coverage-summary self-test	Passed	2026-10-15
Local full test suite	118 passed, 0 failed	2026-10-15
Local format verification	Passed	2026-10-15
Armand human review	Pending	—
Hosted CI run	Pending	—

6. Quality Strategy

6.1 Risk-based test selection

Per `docs/Milestone 3/SEN381_M3_Work_Split.md` and `docs/CivicConnect_M3_Issues.md` (M3-TEST-02).

Top risks prioritized by likelihood × impact:

Risk	Priority	Verification approach	Test cases
Authentication bypass (DEF-003/004)	High	Integration tests with auth boundary	TC-I-03, TC-N-01, TC-N-02
Illegal state transition	High	State transition tests	TC-U-03a/b, TC-I-05, TC-24-05..07
Data persistence failures	High	Integration tests on PostgreSQL	TC-I-01, TC-I-02
Authorization bypass	Medium	Negative/unauthorized suite	TC-N-01..03
Observer failure isolation	Medium	Moq-based failure tests	TC-U-05 (DEF-006)
Dashboard filter correctness	Medium	Count reconciliation tests	TC-U-06
Performance regression	Medium	Load test on dashboard	TC-P-01

6.2 Test allocation

Category	Lead	Technique	Test classes (17)	Count
Unit/component	All	EP/BVA, state transition, Moq	`DomainEntityTests`, `RequestTransitionPolicyTests`, `RequestStateNotifierTests`, `CategoryValidationContextTests`, `UserStoreTests`	67
Black-box functional	Ditheto	EP/BVA, state transition, decision table	`RequesterSliceTests`	15
API/integration	Armand	Success + negative cases	`RequestControllerTests`, `AuthServiceTests`, `PrivilegedProvisioningTests`	51
Negative/unauthorized	Armand	Role-based, ownership, scope	Subset of `RequestControllerTests`, `AuthServiceTests`	13
E2E/system	Riegardt	Critical journey on staging	`E2EJourneyTests`	2
Performance	Riegardt	NBomber load test	`CivicConnect.Perf`	1
Total				**118**

6.3 Meaningful Test Rule

A test counts only if it has:

1. Traceable basis (requirement/risk)
2. Purpose (technique)
3. Deliberate input data
4. Defined expected result
5. Real assertion

6. Execution evidence
7. Interpretation

6.4 Evidence sources

Test evidence file	Trace
`docs/Tests/requester-slice.md`	TC-U-05/06 (Requester slice, 2026-10-15)
`docs/Tests/github_issues_22-25_test_evidence.md`	TC-22-01..25-06 (Staff slice, 2026-10-13)
`docs/Tests/issues_48_53_56_test_evidence.md`	TC-U-03..04, TC-I-01..05, TC-N-01..03 (2026-10-14)
`docs/Tests/m3-test-05-evidence.md`	Notifiers + dashboard (2026-10-15)
`docs/Tests/issue_27_observer_test_evidence.md`	Observer fan-out + failure isolation (2026-10-14)

7. Black-Box, Integration, and System Tests

7.1 Unit tests (TC-U-03..06)

From `docs/Tests/issues_48_53_56_test_evidence.md` and `docs/Tests/m3-test-05-evidence.md`:

ID	Basis	Result
TC-U-03a	State transition Open→Assigned allowed	PASS
TC-U-03b	State transition Closed→Open rejected	PASS
TC-U-04	Role policy: requester cannot reject	PASS
TC-U-05	Observer fan-out, failure isolation	8/8 PASS
TC-U-06	Dashboard counts + filters	3/3 PASS

7.2 Black-box functional (TC-B)

Technique	Basis	Cases
Equivalence Partitioning / BVA	FR-001 submission fields	TC-B-01..04 (Ditheto, requester slice)
State Transition Testing	FR-011 status changes	TC-B-05..07 (legal/illegal transitions)
Decision Table	NFR-018 role×action	TC-B-08..10 (who can do what)

7.3 API/integration tests

ID	Basis	Input	Expected	Actual	Status
TC-I-01	FR-001 submission + initial history	Valid category-1 request, authenticated requester	201 DTO, one Open request, one linked history row	Confirmed	PASS (local)
TC-I-02	FR-001 unknown category	CategoryId = MaxValue	400, no persistence	Confirmed	PASS (local)
TC-I-03	NFR-017 anonymous	Anonymous POST /api/requests	401, no request	Confirmed	PASS (local)
TC-I-04	FR-009 + FR-012	Assigned→InProgress, observer	200, exactly one history row, observer notified	Confirmed	PASS (local)
TC-I-05	FR-009 illegal transition	Open→Resolved	400, state unchanged, no history	Confirmed	PASS (local)

CI must be green against PostgreSQL for TC-I-01 as complete relational evidence (InMemory does not enforce FKs).

7.4 Negative/unauthorized tests

ID	Basis	Input	Expected	Actual	Status
TC-N-01	NFR-018	Requester calls GET /api/staff/requests	403	Confirmed	PASS
TC-N-02	NFR-019	Requester A reads requester B's request	403 or 404	403, target still exists	PASS
TC-N-03	FR-013, NFR-019	Staff B reads request assigned to staff A	403, unchanged	Confirmed	PASS

7.5 E2E journeys (TC-S-01, TC-S-02)

From `src/CivicConnect.Tests/E2EJourneyTests.cs`:

ID	Journey	Evidence
TC-S-01	Submit → staff assigns → resolves → requester sees update + history at DB level	`SubmitAssignResolve_RequesterSeesUpdate` — 200/0 history rows match expected states
TC-S-02	Resolved request → feedback → dashboard reflects it	`ResolvedFeedback_DashboardReflectsIt` — 201 feedback, 200 dashboard, total=1, byState Resolved=1

Staging run (M3-TEST-15, Issue #59): `STAGING-E2E PASS: TC-S-01 green on commit ec9cb7a` via `scripts/staging-e2e.ps1`.

7.6 Postman test cases

`docs/Postman_Request_Comments_Test_Cases.md` — 14 manual HTTP API test cases (TC-01..14) covering comment creation, visibility, ownership, authorization, and error paths.

8. Performance and Reliability

8.1 Performance operation

From `docs/Riegardt-Decisions.md` M3-PERF-01 (Issue #61, decided by Riegardt):

Parameter	Value
Operation	`GET /api/reports/requests-summary` (filtered + unfiltered)
Tool	NBomber 6.7.0
Virtual users	10
Duration	2 minutes steady state
Data	1,000 seeded requests
Target	p95 < 3s, zero errors
Environment	Local app + isolated local PostgreSQL (shared DB untouched)

8.2 Result (TC-P-01, M3-PERF-03 #63)

Run: 2026-10-15, isolated scratch cluster

Metric	Value
Requests	109,525
Errors	0
RPS	~913
p50	9.45ms
p75	14.27ms
p95	**23.71ms**
p99	32.9ms
Max	486ms

Verdict: PASS — p95 23.71ms << 3,000ms target, zero errors (~100× headroom)

8.3 Limitations

- Production behavior: not proven (different host, volume, network)
- Concurrent-write contention: not tested (read-only workload)
- Beyond 10-user scale: not tested (steady-state measurement)

- Cold-start/JIT warm-up: excluded
- NBomber free license: personal-use-only; flag before organizational use

8.4 Reliability

- `/health` (`M3-STG-07`, Issue #81): 200 Healthy with DB connected; 503 with generic message on DB-down, no credential/host leakage
- Rollback/recovery (`M3-STG-09`, Issue #83): Forward-fix policy; rollback requires prior `pg_dump` backup; `Down()` drops `AuditLogs` table (5 rows lost in dry run)

8.5 Environment differences (M3-STG-08, Issue #82)

Area	Dev	Staging (interim)	Staging (compose)	Intended production
Hosting	`dotnet run`	Same (`ASPNETCORE_ENVIRONMENT=Staging`)	Containers	Undecided.
Database	Local PG18 or shared remote	Separate scratch DB	Compose volume	Remote DB.
Data volume	Handful + 1k perf seed	Same	Same	0-10k records.
Swagger	On	On (explicit decision)	On	Off
Secrets	`.env` (gitignored)	Env vars	Same via compose	Commited on Deploy server under encryption.
TLS	None	None	None	Expected (HSTS)
Auth	TestaAuth minting	Same via throwaway minter	Same	JWT Bearer token and bcrypt password hashing.
Build identity	Assembly version	`GIT_SHA` env	Sme	Tag-based

9. Defects and Quality Interpretation

9.1 Canonical findings register

`docs/CI/M3-CI-07-findings.md` (see Section 5.6 for the table). Findings disposition:

Finding	Interpretation	Status	
FIND-001..004, 006, 007	Fixed or mitigated for stated scope	Closed for recorded scope; re-verify in CI	
FIND-005	No vulnerable packages on 2026-10-14	Accepted with monitoring by Dependabot (RR-SEC-02)	
FIND-008 / DEF-005	Historical credential literal in Git history	Mitigated, open (RR-SEC-01, TD-SEC-01)	
FIND-009	Unrelated .gitignore change	Out of scope	Accepted
FIND-010	Hosted GitHub settings not enabled	Mitigated (RR-SEC-03)	

9.2 Defect Register

See `docs/CivicConnect_M3_Issues.md` (M3-DEF-01) and source analysis. Key defects:

ID	Defect	Status
DEF-001	Services not registered in DI	Fixed (PR #121/#123)
DEF-002	StatusHistory linked before key exists	Fixed (navigation-based linking)
DEF-003	UpdateState accepts any transition	Fixed (`RequestStateTransitionPolicy`)
DEF-004	Hardcoded identities	Fixed (JWT + role attributes)
DEF-005	DB password in Git history	Mitigated (current value empty for example reasons)
DEF-006	Observer failure after SaveChanges	Fixed (failure isolation in `RequestStateNotifier`)

9.3 Requirement traceability

`docs/RTM-M3.md` is authoritative for requirement wording, acceptance criteria, status vocabulary, and evidence limits. The matrix continues M1/M2 baseline requirements across requester (FR-001–006), staff/security (FR-007–013, NFR-017–019), and management (FR-014–016, NFR-020, NFR-024) slices. Status claims below reflect RTM evidence, not coverage percentages.

Requirement	Current evidence and disposition
FR-001–002	**Implemented.** Request submission validation, category selection, category-1 strategy and fallback; see `RequesterSliceTests` and `docs/Tests/requester-slice.md`.
FR-003	**Verified** against accepted numeric `Request.Id` reference decision; `CreateRequest_ValidSubmission_ReturnsCreatedDtoAndPersistsInitialHistory`.
FR-004–005	**Verified in test host.** Request ownership, paging/state projection, history chronology/ownership; `RequesterSliceTests.GetMyRequests_*`, `GetHistory_IncludesInitialAndChronologicallyOrdersRows_AndEnforcesOwnership`.
FR-006	**Implemented.** Owner/completion/rating/comment/duplicate constraints; `RequesterSliceTests.Feedback_*`; PostgreSQL unique-index enforcement still depends on relational verification.
FR-007	**Implemented.** Role-gated staff queue and request-detail access have evidence (`GetStaffRequests_Requester_ReturnsForbidden`, `GetRequest_OtherRequester_ReturnsForbidden`, `GetRequest_StaffOutsideAssignment_ReturnsForbidden`), but RTM correctly leaves staff visibility/scope incomplete: queue returns all requests to STAFF/MANAGER and `AccessControlService` is not applied there.
FR-008	**Verified for implemented filters/paging only.** `GetStaffRequests_CombinesFiltersAndPaging`; request list filters state/category/assignee and orders by creation time, but search is not implemented.
FR-009	**Implemented.** Detail endpoint includes category/comments/feedback, ownership checks, and requester internal-comment filtering. `GetRequest_SeededRequestIncludesComment` and requester/staff visibility tests prove selected fields/behavior, not every baseline detail attribute.
FR-010	**Verified for covered endpoint behavior.** Assignment writes assignee/state/history/audit and rejects requester role; `AssignRequest_UpdatesStateHistoryAndAuditWithActor`, `AssignRequest_Requester_ReturnsForbiddenAndLeavesRequestOpen`.
FR-011	**Implemented.** Transition policy and legal/illegal behavior are tested (`RequestTransitionPolicyTests`, `UpdateState_ClosedToOpen_ReturnsBadRequest`, `UpdateState_IllegalTransition_LeavesRequestUnchanged`, reason tests). Full transition/role matrix and exactly-one history/audit verification remain incomplete.

Requirement	Current evidence and disposition
FR-012	**Implemented.** Public/internal comments and visibility are tested (`AddComment_*`, `GetRequest_StaffCanSeeInternalComments`, `GetRequest_RequesterCannotSeeInternalComments`). Dedicated resolution-information assertion remains absent; observer tests prove best-effort notification only.
FR-013	**Implemented.** State transitions to Resolved/Closed exist, and `E2EJourneyTests.SubmitAssignResolve_RequesterSeesUpdate` verifies requester can retrieve the resolved request plus DB history. Journey does not assert response state/history through requester DTO/endpoint, so full requester-visible resolution criterion remains incomplete.
NFR-017	**Implemented; production-like JWT path partly verified.** `JwtAuthFlowTests.RegisterCreateListRead_WithRealJwt_PreservesOwnershipAndRoleRestrictions` exercises registration, issued JWT, ownership, role restriction, and login through JWT bearer. It does not independently validate malformed/expired/wrong-issuer/wrong-audience/signature cases or a real external IdP.
NFR-018	**Implemented; test-host verified.** Role gates and denial/no-side-effect cases cover requester vs staff/manager actions. Test-header authentication proves authorization wiring; JWT flow test covers a real app-issued token for requester flow.
NFR-019	**Implemented.** Requester ownership and internal-comment visibility are evidenced, and staff detail outside-assignment denial is now directly tested by `GetRequest_StaffOutsideAssignment_ReturnsForbidden`. However, `GetStaffRequests` is not assignment-scoped, so the complete staff object-level scope requirement is not met.
FR-014–016, NFR-020, NFR-024	Management RTM records verification for dashboard filters/counts, stalled/reassignment, category administration, audit reads/writes, and stated localhost performance workload. See `DashboardSummaryTests`, `ManagementTests`, `AuditLogTests`, TC-P-01, and `docs/RTM-M3.md` management rows. FR-015/016 acceptance criteria are authored by Riegardt; dashboard age buckets deferred.

RTM evidence limits remain material: current evidence mixes local automated tests, historical manual HTTP records, test-host checks, and live shared-database checks. `docs/Tests/github_issues_22-25_test_evidence.md` labels several outcomes as recorded/manual and explicitly notes missing state/no-write follow-up evidence. `docs/Tests/issues_48_53_56_test_evidence.md` also describes an earlier 75-test run; neither historical count supersedes the current run below. Full legal transition permutations, requester-visible resolved state assertions, exhaustive detail-field assertions, and production-like JWT negative cases remain gaps. Do not elevate **In Development** rows to **Verified** without new criterion-matched evidence and RTM review.

9.4 Quality gates evidence

The checked-in workflow `.github/workflows/test.yml` defines frontend build, .NET restore/build, PostgreSQL migration and schema check, formatting, NuGet vulnerability scan, coverage-script self-test, full test/coverage collection, coverage summary, and TRX/Cobertura artifact upload. Its presence proves configured gates, not that this exact checkout passed hosted CI. No hosted run URL or hosted artifact was verified for this document update.

Verification	Evidence/result	What it establishes / does not establish
Full local tests	2026-10-16, `dotnet test src/CivicConnect.slnx --configuration Release --no-restore --logger "console;verbosity=minimal"`: **118 passed, 0 failed, 0 skipped**	Current local test host result; local factory uses EF Core InMemory unless `ConnectionStrings__DefaultConnection` is supplied. It is not PostgreSQL/hosted CI evidence.
Real JWT flow	Included in the 118-test run: `JwtAuthFlowTests.RegisterCreateListRead_WithRealJwt_PreservesOwnershipAndRoleRestrictions`	Exercises app-issued JWT through bearer handler and requester access/role path; does not cover token-validation negative matrix or external IdP.
Format/vulnerability/coverage gates	Commands are configured in workflow; earlier local results and self-test are recorded in Section 5.7 / CI evidence docs	Do not claim a fresh local result for this update. Vulnerability findings are scan-time dependent; see `docs/CI/M3-CI-07-findings.md` FIND-005 and RR-SEC-02.

Verification	Evidence/result	What it establishes / does not establish
PostgreSQL/migrations	CI workflow configures disposable PostgreSQL, applies migrations, confirms `Requests` table; `DatabaseConnectionTests` passed in the 118-test local run	Connectivity to a configured PostgreSQL database passed; this alone does not prove migrations or that all API tests used PostgreSQL (`TestWebApplicationFactory` switches providers only when `ConnectionStrings__DefaultConnection` is set).
Hosted CI and GitHub settings	Pending current-run URL/artifacts and live settings evidence	Local checks cannot establish hosted result, required status checks, Dependabot/secret-scanning settings, or repository protection state; see FIND-009, RR-SEC-03.

Coverage interpretation and critical-path limits are documented in `docs/Milestone 3/PED-Section-5-CI-and-Coverage.md`: line/branch execution is not requirement acceptance evidence. Keep the historical 59/75/93-test records date- and scope-labelled; current local result is 118, not a replacement for the missing hosted run.

9.5 Review record

AI assistance and review responsibility are recorded in the role-specific registers: `docs/Ditheto-AI-Usage-Register.md`, `docs/Armand-AI-Usage-Register.md`, and `docs/Riegardt-AI-Usage-Register.md`. Ditheto/Armand registers explicitly leave human review pending for their listed artefacts; Riegardt register records Riegardt's own confirmed reviews for its listed work. These entries are not blanket approval of this consolidated PED or of other owners' requirement mappings.

For this update, RTM evidence was cross-checked against current test names, `RequestsController`, `AccessControlService`, `Program.cs`, workflow configuration, and evidence-folder caveats. This was an AI-assisted repository consistency review, not independent human sign-off. Ditheto/Armand owners must verify their RTM mappings, acceptance criteria, citations, and status changes; release owner must reconcile management claims and dated evidence. Record completed checks with reviewer, date, and exact artefacts in the corresponding AI/review register. Until then, human verification remains pending; hosted-CI and production-readiness claims remain unverified.

10. Release Readiness

10.1 Evidence summary

`docs/Release-Evidence-Summary.md` (M3-REL-01, #98). Nine areas:

Area	Result	Interpretation
Build	0 warnings, 0 errors (analyzer gate)	Compiles under warnings-as-errors
CI	Workflow green on phase PRs	Pipeline enforces gates
Regression	DEF-006 test; Closed→Open rejection; illegal-transition suite	Fixed defects stay fixed
API/integration	All controller tests green (auth, RBAC, transitions, comments, assign, queue)	Endpoints enforce roles, ownership, policy
E2E/system	TC-S-01 + TC-S-02 green	Critical journeys work locally and on staging
Static/security	Analyzer gate clean; no vulnerable packages	No open findings; .editorconfig exceptions mitigated
Performance	TC-P-01: 109,525 reqs, 0 errors, p95 23.71ms	~100× headroom
Defects	DEF-006 verified fixed; pre-existing issues fixed	No known open Riegardt-area defects
Acceptance	RTM rows verified	Acceptable per implemented contract

10.2 Readiness classification

CONDITIONALLY READY (M3-REL-02, Issue #99, Riegardt judgment call)

For READY:

- Clean build under warnings-as-errors
- 118/118 tests including E2E journeys locally and on staging
- Performance PASS with ~100× headroom
- Static/security gates clean
- Audit trail writes verified
- Full evidence indexed

Against READY (residual conditions):

- No production host, TLS story, or real IdP (JWT demo scheme)
- Staging is local-simulated
- Dual audit tables unresolved (TD-04)
- FR-015/016 AC self-authored by Riegardt (no M1 baseline)
- No concurrency token (TD-02)
- Production JWT validation not exercised by tests (RR-SEC-04)

10.3 Readiness conditions

1. **Demo/assessment use only** via documented paths (local run, Swagger, staging scripts)
2. **No production traffic** until: real IdP, TLS/hosting story, one-table audit decision
3. **Deferred scope stays deferred** (age buckets, outbox, FR-015/016 baseline reconciliation)

10.4 Residual risk summary

From `docs/Residual-Risk-Statement.md` (M3-REL-03, #100):

ID	Risk	Mitigation	Status	Owner
RR-SEC-01	Historical DB password in Git history	Current value empty; CI uses disposable credentials	Resolved	Armand/repository owner
RR-SEC-02	NuGet advisories time-varying	CI rescans every run	Accepted/monitoring	Armand
RR-SEC-03	Hosted GitHub security settings not enabled	Repository evidence recorded; Security settings saved on pdf	Resolved	Armand/repository owner
RR-SEC-04	Test auth doesn't exercise JWT validation	Production JWT config present; documented limitation	Resolved	Armand
RR-REL-01	No production host/TLS/backup	Staging defined; demo only	Open	Riegardt

ID	Risk	Mitigation	Status	Owner
RR-REL-02	No real identity provider	JWT real where exercised	Resolved	Riegardt
RR-REL-03	Dual audit tables (AuditEntries vs AuditLogs)	One-table team decision pending	Open	Riegardt + Armand
RR-REL-04	Performance claim is localhost-only	Limits documented	Accepted	Riegardt
RR-REL-05	`docker compose up` not executed	Dockerfile/compose reviewed; needs Docker host	Resolved(Gets hosted in CI)	Riegardt
RR-REL-06	FR-015/016 AC self-authored	Traceability recorded	Resolved	Riegardt

10.5 Technical debt

From `docs/Technical-Debt-Register.md` (M3-REL-04, #101):

ID	Item	Impact	Interest	Status	Owner
TD-01	Repository abstraction bypassed	Testability + layering	Grows per endpoint	Open, pending ADR	Ditheto
TD-02	No concurrency token on Request	Last-writer-wins	Grows with staff usage	Open	Armand
TD-03	Observer not outbox-based	Lost notifications	Grows if relied upon	Accepted	Riegardt
TD-04	Dual audit tables	Split evidence	Blocks audit wiring	Open	Riegardt + Armand
TD-05	Dead authorization path	Divergent role strings	Next endpoint may copy	Open	Armand
TD-06	TestAuth crashes on missing JWT_SECRET	Broken local demo flow	Hits new contributors	Open	Armand
TD-07	Audit read capped at Take(200)	Unusable at scale	Grows with audit volume	Accepted	Riegardt
TD-08	Swagger only in Development	Assessors can't explore staging	Blocks demo	Open	Riegardt
TD-09	Dead migration file	DB confusion	Static	Open	Armand
TD-10	Undefined deferred scope	Can't plan	Grows if scope creeps	Open	All
TD-11	UseExceptionHandler points at missing controller	500s as framework pages	Until Phase 2 PR	Pending merge	Riegardt

11. Risk Register

Combined from Ditheto (quality/requirements), Armand (security/process), Riegardt (release/operations) perspectives, per M3-RISK-01..03.

ID	Risk	Category	Probability	Impact	Mitigation	Status	Owner
R1	CI slips and removes everyone's gate	Process	Medium	High	CI started in Phase 1; quality gate blocks merge	Open	All
R2	Migration snapshot conflicts	Process	Medium	Medium	M3-GOV-04 migration coordination rule	Open	Armand
R3	Auth rework late	Security	Low	High	JWT implemented in Armand slice; test limitations documented (RR-SEC-04)	Open	Armand
R4	Secrets in history	Security	Medium	High	Current config empty; historical credential in Git (DEF-005/FIND-008); rotation pending owner action	Open	Armand/owner
R5	Requirement-implementation drift	Quality	Medium	High	RTM updated in same PRs as tests	Open	Ditheto
R6	Late RTM assembly	Quality	High	Medium	Updated per-test as tests land	Open	Ditheto
R7	Low-value tests counted as padding	Quality	Medium	Medium	Meaningful Test Rule enforced	Open	Ditheto
R8	Staff scope restriction not implemented	Security	High	High	`GetRequest_StaffOutsideAssignment_ReturnsForbidden` tested; `GetRequest_OtherRequester_ReturnsForbidden` tested; assigned staff may still access out-of-scope requests — documented gap	Open	Armand
R9	Production JWT not exercised	Security	High	High	Test handler proves wiring only (RR-SEC-04); production JWT validation not tested	Open	Armand
R10	No production hosting/TLS/IdP	Release	High	High	Conditional READY; demo/assessment only	Open	Riegardt
R11	Dual audit tables unresolved	Release	Low	Medium	One-table decision pending (TD-04/RR-REL-03)	Open	Riegardt
R12	Performance claim is localhost-only	Release	Medium	Medium	Limits documented in Riegardt-Decisions.md	Accepted	Riegardt

12. Evidence Index

12.1 Core registers

Register	File	Owner
Requirements Traceability Matrix	`docs/RTM-M3.md`	Ditheto (continues with Armand/Riegardt rows)

Register	File	Owner
Defect Register	see `docs/CivicConnect_M3_Issues.md` (DEF-nnn)	Ditheto
Technical Debt Register	`docs/Technical-Debt-Register.md`	Riegardt
Residual Risk Statement	`docs/Residual-Risk-Statement.md`	Riegardt
Static/security findings	`docs/CI/M3-CI-07-findings.md`	Armand
AI Usage Register (Ditheto)	`docs/P1-AI-Usage-Register.md`	Ditheto
AI Usage Register (Armand)	`docs/P2-AI-Usage-Register.md`	Armand
AI Usage Register (Riegardt)	`docs/P3-AI-Usage-Register.md`	Riegardt
Riegardt Decisions log	`docs/P3-Decisions.md`	Riegardt
Release Evidence Summary	`docs/Release-Evidence-Summary.md`	Riegardt

12.2 Test evidence

File	Trace
`docs/Tests/requester-slice.md`	FR-001–006, TC-U-05/06
`docs/Tests/github_issues_22-25_test_evidence.md`	FR-007–013, TC-22..25
`docs/Tests/issues_48_53_56_test_evidence.md`	TC-U-03..04, TC-I-01..05, TC-N-01..03
`docs/Tests/m3-test-05-evidence.md`	TC-U-05, TC-U-06
`docs/Tests/issue_27_observer_test_evidence.md`	FR-012, DEF-006, TC-I-04
`docs/Postman_Request_Comments_Test_Cases.md`	TC-01..14 (manual HTTP)

12.3 Source evidence

Artefact	Path
Solution file	`src/CivicConnect.slnx`
Controllers	`src/CivicConnect.Web/Controllers/`
DbContext + migrations	`src/CivicConnect.Web/Data/`, `src/CivicConnect.Web/Migrations/`
DB schema script	`src/CivicConnect.Web/db_schema.sql`
State transition policy	`src/CivicConnect.Application/Policies/RequestStateTransitionPolicy.cs`
Observer pattern	`src/CivicConnect.Application/Patterns/Observer/`
Strategy pattern	`src/CivicConnect.Application/Patterns/Strategy/`

Artefact	Path
CI workflow	`.github/workflows/test.yml`
Dockerfile	`Dockerfile`
Staging compose	`docker-compose.staging.yml`
Vulnerability gate script	`scripts/check-vulnerabilities.py`
Coverage script	`scripts/coverage-summary.py`
Staging E2E script	`scripts/staging-e2e.ps1`
Staging smoke script	`scripts/staging-smoke.ps1`
Performance harness	`src/CivicConnect.Perf/`
Perf seed	`src/CivicConnect.Perf/seed-perf.sql`

12.4 PED internal cross-references

- Section 2 ⟷ Section 3 (conformance deviations → controlled changes)
- Section 5 ⟷ Section 7 (CI evidence → test results)
- Section 5 ⟷ Section 9 (CI findings register → defect discharge)
- Section 6 ⟷ RTM (`docs/RTM-M3.md`) (risk-based selection trace)
- Section 8 ⟷ Section 9 (perf limits → interpretation)
- Section 8 ⟷ Section 10 (perf evidence → readiness)
- Section 10 ⟷ Residual Risk Statement (RR-REL-01..06)
- Section 10 ⟷ Technical Debt Register (TD-01..TD-11)
- Section 10 ⟷ Release Evidence Summary (M3-REL-01)

13. Harvard References and Citations

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Appendix A: M2 Baseline Version History

Version	Date	Changes
PED v1.0	M1	Initial baseline (PED v1.0) — milestone brief
PED v2.0	M2	Full PED with FR-001–016, NFR-017–020, NFR-024; layered architecture; Observer/Strategy patterns; REST API; PostgreSQL/EF Core; React SPA baseline
PED v3.0	M3	Continued from v2.0 with M3 implementation evidence, controlled changes, CI gates, test results, staging, performance, and release readiness

Appendix B: Controlled Changes Log

Change	Original	Revised	Evidence	Status
Repository abstraction	Repository interfaces	DbContext-as-UoW accepted	TD-01	Accepted
API versioning	`/api/v1/`	`/api`	RTM evidence limits	Accepted
State pattern	State objects	Central transition table	`RequestStateTransitionPolicy.cs`	Accepted
Auth mechanism	Identity provider TBD	JWT bearer + test handler	`Program.cs`, RR-SEC-04	Accepted with risk
Frontend	React 19 SPA	React 19 SPA (merged PR #134)	`ClientApp/`, Dockerfile	Conformant
Notification	Synchronous observer	Log-only, best-effort (observer kept)	`RequestStateNotifier.cs`, TD-03	Accepted
Concurrency	None specified	None (no token)	TD-02	Open